Grado: 8 **Grupo: F**

Materia: Diseño Electrónico Basado en Sistemas Embebidos

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Código fuente:

```
#include <Arduino.h>

#include <WiFi.h>

#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#include <Adafruit_BMP280.h>

#include <DHT.h>

#include "ThingSpeak.h"


/*ADVERTENCIA:

  Utiliza una fuente adicional de 3.3v para alimentar los sensores, al utilizar
WiFi la ESP32 tiene picos de

  voltaje que pueden causar el reinicio de la ESP32.

  Solo asegurate de conectar el GND de la ESP32 al GND de la fuente de
3.3v*/


// --- CONFIGURACIÓN ---

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const char* SSID = "UAM-ROBOTICA";

const char* PASS = "m4nt32024uat";

unsigned long CHANNEL_ID = 3284944;

const char* WRITE_API_KEY = "JOLB193MIUOZ402S";


#define DHTPIN 18       // Pin de datos del DHT22
```
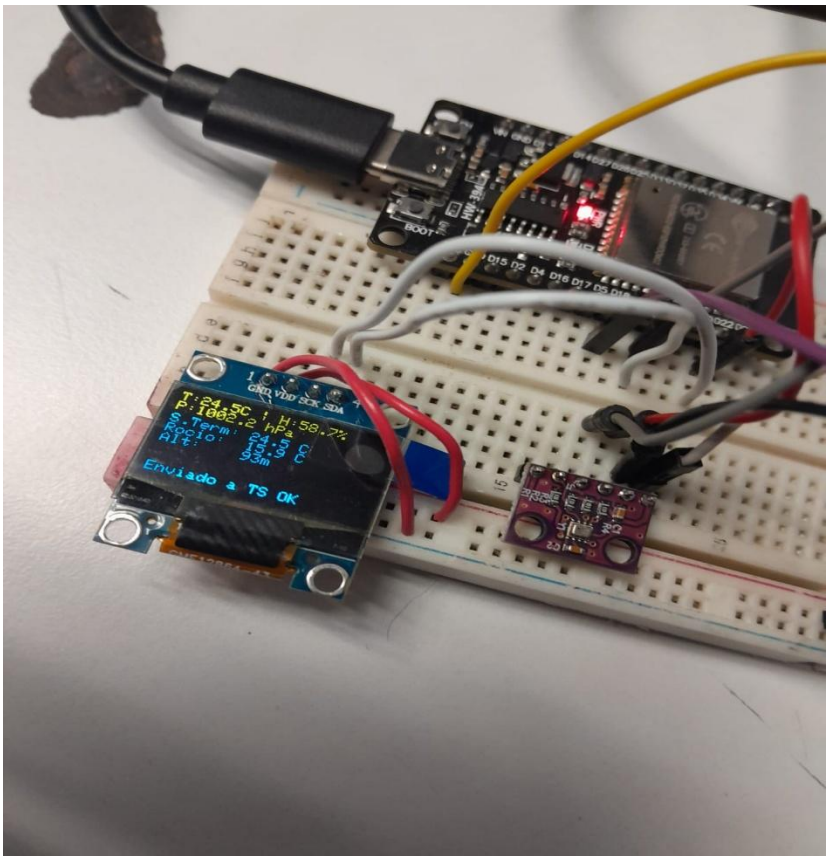
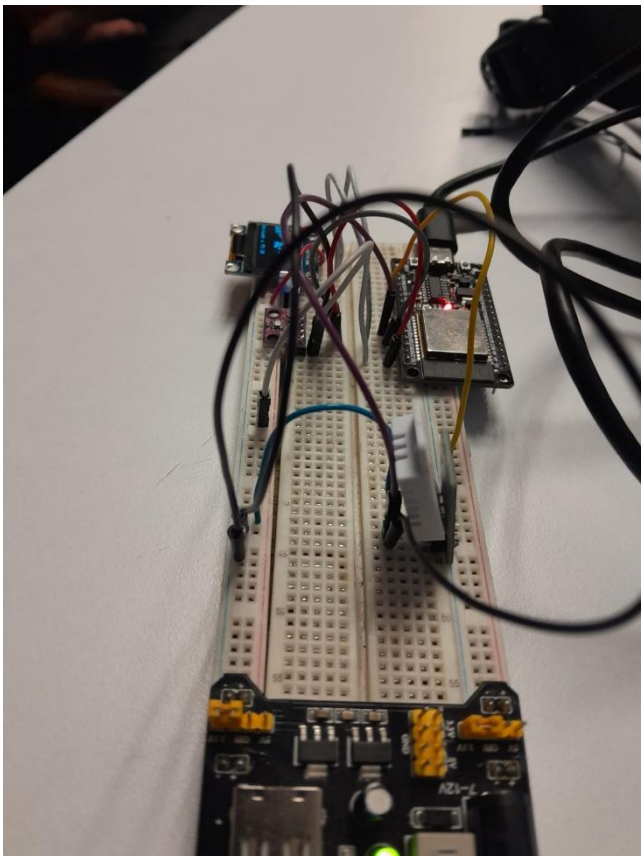

```
#define DHTTYPE DHT22

// --- INSTANCIAS ---
Adafruit_SSD1306 display(128, 64, &Wire, -1);
Adafruit_BMP280 bmp;
DHT dht(DHTPIN, DHTTYPE);
WiFiClient client;

void setup() {
  Serial.begin(115200);
  dht.begin();

  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    for(;;);
  }
  display.clearDisplay();
  display.setTextColor(WHITE);

  if (!bmp.begin(0x76)) {
    Serial.println("Error BMP280");
    while (1);
  }

  WiFi.begin(SSID, PASS);
```

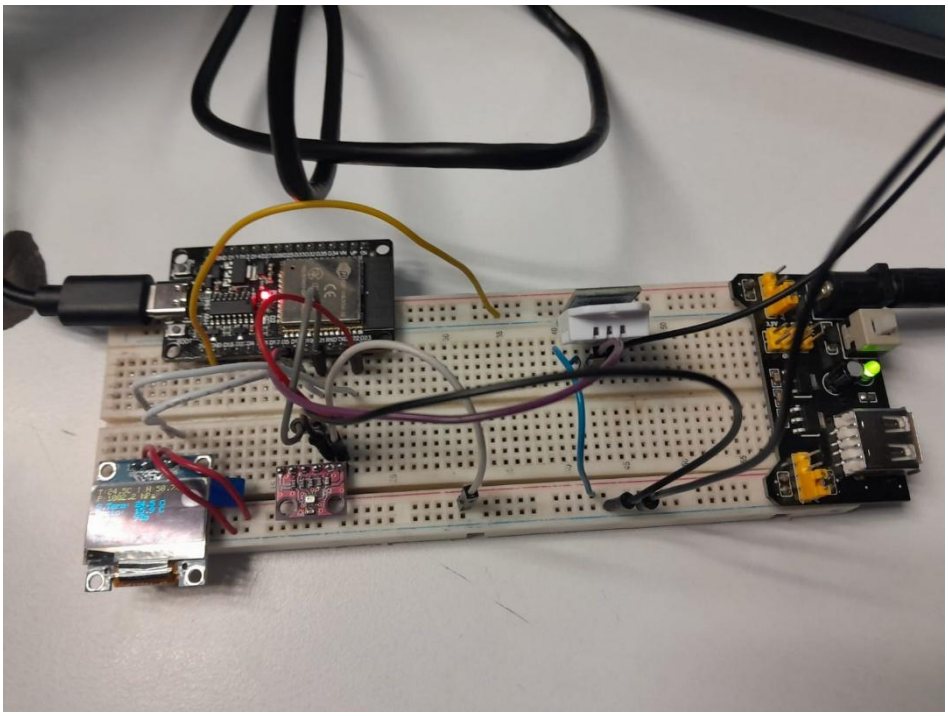
```
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  ThingSpeak.begin(client);
}

void loop() {
  // Lecturas de sensores
  float temp_bmp = bmp.readTemperature();
  float pres = bmp.readPressure() / 100.0;
  float alt  = bmp.readAltitude(1013.25);

  float hum = dht.readHumidity();
  float temp_dht = dht.readTemperature();

  // Validar lectura DHT
  if (isnan(hum) || isnan(temp_dht)) {
    Serial.println("Error leyendo DHT22");
    return;
  }

  // Usamos la temperatura del BMP280 por ser más precisa, pero la
humedad del DHT
```

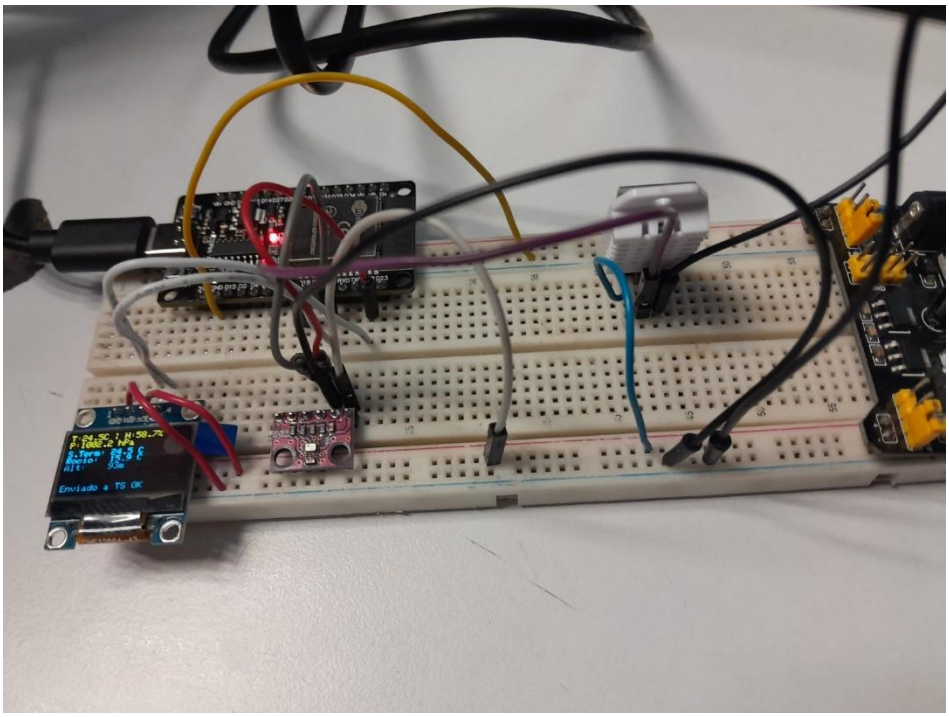
```
// 1. Sensación Térmica (Heat Index oficial de la NOAA)
float feelsLike = dht.computeHeatIndex(temp_bmp, hum, false);

// 2. Punto de Rocío (Fórmula de Magnus)
float a = 17.27; float b = 237.7;
float gamma = ((a * temp_bmp) / (b + temp_bmp)) + log(hum/100.0);
float dewPoint = (b * gamma) / (a - gamma);

// --- ENVIAR A THINGSPEAK (8 CAMPOS) ---
ThingSpeak.setField(1, temp_bmp);   // Temp BMP280
ThingSpeak.setField(2, hum);        // Humedad Real DHT22
ThingSpeak.setField(3, pres);       // Presión
ThingSpeak.setField(4, feelsLike);  // Sensación Térmica
ThingSpeak.setField(5, dewPoint);   // Punto de Rocío
ThingSpeak.setField(6, alt);        // Altitud
ThingSpeak.setField(7, WiFi.RSSI());// Señal WiFi
ThingSpeak.setField(8, temp_dht);   // Temp DHT22 (para comparar)

int status = ThingSpeak.writeFields(CHANNEL_ID, WRITE_API_KEY);

// --- MOSTRAR EN OLED ---
display.clearDisplay();
display.setTextSize(1);
display.setCursor(0,0);
```

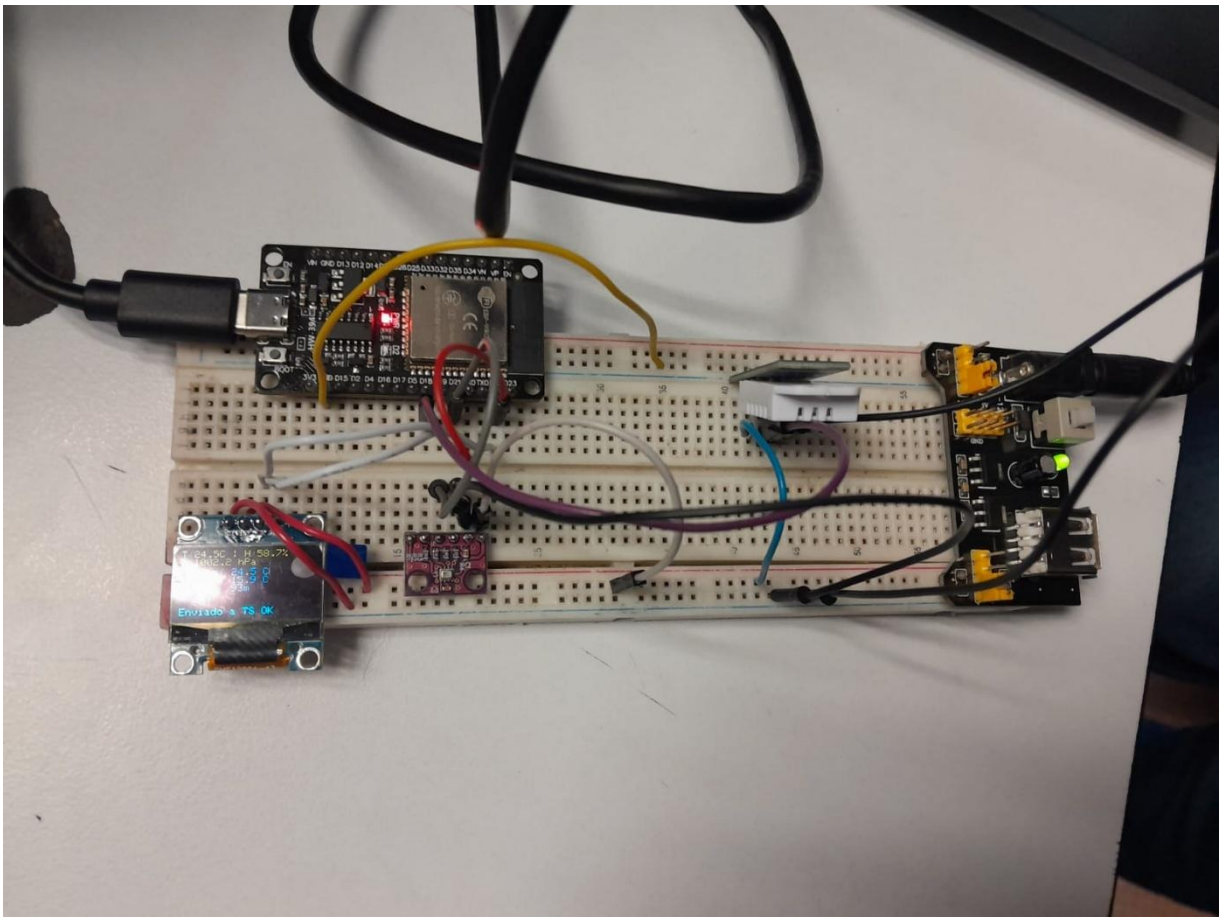
```
  display.printf("T:%.1fC | H:%.1f%%\n", temp_bmp, hum);
  display.printf("P:%.1f hPa\n", pres);
  display.printf("S.Term: %.1f C\n", feelsLike);
  display.printf("Rocio:  %.1f C\n", dewPoint);
  display.printf("Alt:    %.0fm\n", alt);

  display.setCursor(0, 55);
  if(status == 200) display.print("Enviado a TS OK");
  else display.print("Error TS: " + String(status));

  display.display();

  delay(20000); // Espera de 20s para ThingSpeak
}
```

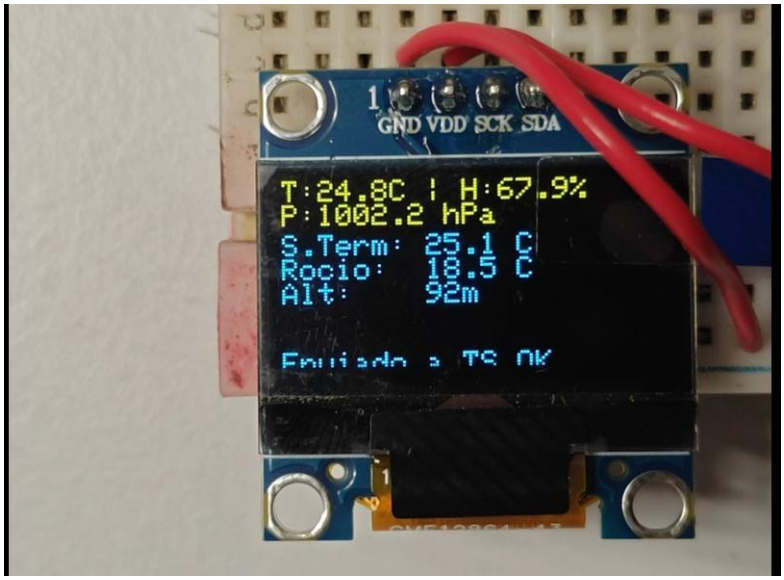

T:24.5C | H:58.7%
P:1002.2 hPa
S.Term: 24.5 C
Rocio: 13.3 C
Alt: 93m
Enviado a TS OK



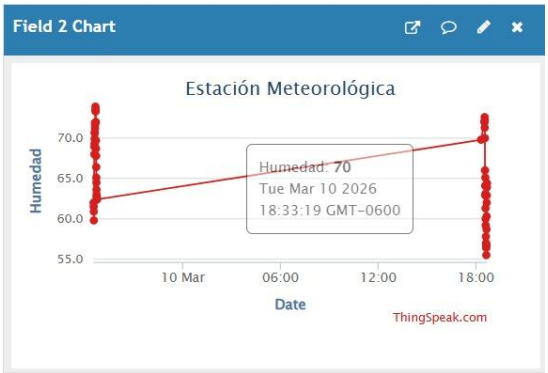


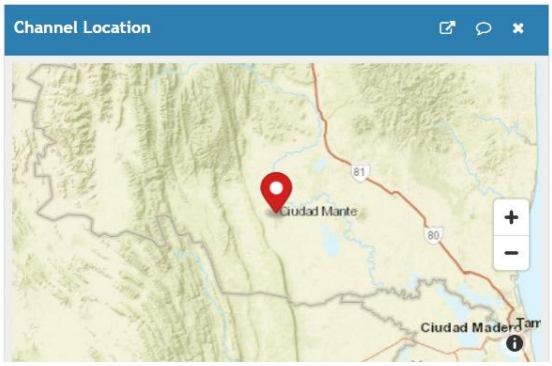

Enviado a TS OK

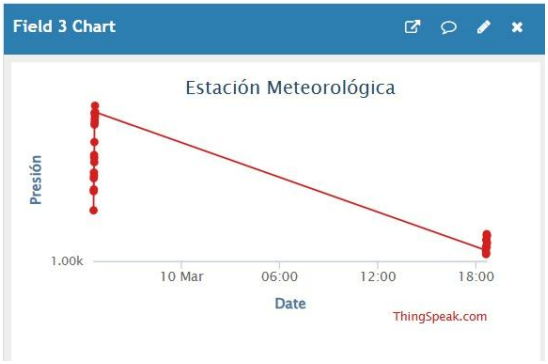


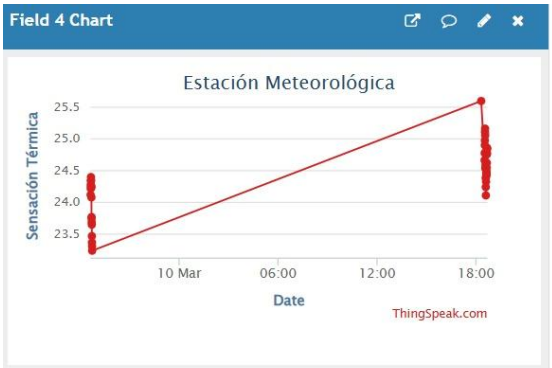

GND VDD SCK SDA
T:24.8C | H:67.9%
P:1002.2 hPa
S.Term: 25.1 C
Rocio: 18.5 C
Alt: 92m

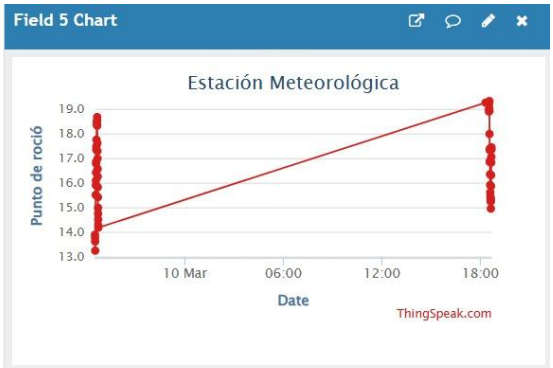
Field 1 Chart
Estación Meteorológica
Temperatura(BMP280)
25.0
24.0
23.0
22.0
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com


Field 2 Chart
Estación Meteorológica
Humedad
70.0
65.0
60.0
55.0
Humedad: 70
Tue Mar 10 2026
18:33:19 GMT-0600
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com


Channel Location
Ciudad Mante
81
80
Ciudad Madero

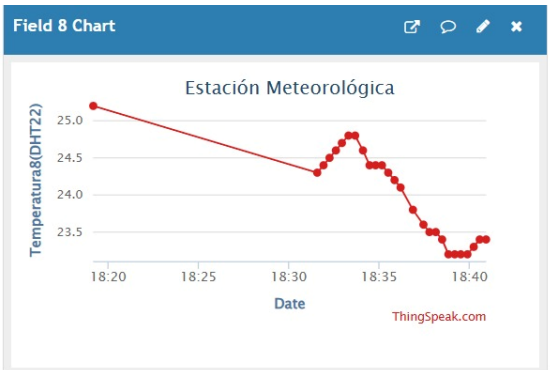

Field 3 Chart
Estación Meteorológica
Presión
1.00k
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com


Field 4 Chart
Estación Meteorológica
Sensación Térmica
25.5
25.0
24.5
24.0
23.5
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com


Field 5 Chart
Estación Meteorológica
Punto de rocío
19.0
18.0
17.0
16.0
15.0
14.0
13.0
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com

Field 6 Chart
Estación Meteorológica
Altitud
80.0
60.0
40.0
20.0
10 Mar
06:00
12:00
18:00
Date
ThingSpeak.com

Field 7 Chart
Estación Meteorológica
Señal wifi
−45.0
−50.0
−55.0
18:20
18:25
18:30
18:35
18:40
Date
ThingSpeak.com


Field 8 Chart
Estación Meteorológica
Temperatura8(DHT22)
25.0
24.5
24.0
23.5
18:20
18:25
18:30
18:35
18:40
Date
ThingSpeak.com